Quantum Cryptography and Secure Drone Communication: A Resilient Autonomous Protocol (RAP) Framework

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***Abstract*—This paper proposes a multi-layered system architecture for forward-secure UAV communication using decoy-state BB84 Quantum Key Distribution (QKD). While QKD offers information-theoretic security, maintaining stable links on mobile platforms remains a challenge. We integrate a two-stage Acquisition, Pointing, and Tracking (APT) system with high-precision clock synchronization to achieve pico second-level timing fidelity. To address operational disruptions, we introduce the Resilient Autonomous Protocol (RAP), a state machine that monitors the Quantum Bit Error Rate (QBER) and secure key rates in real-time. Upon detected eavesdropping, the RAP transitions the UAV to an autonomous ”communications-dark” state, ensuring mission integrity despite quantum channel compromise.**

***Index Terms*—AES 256 cipher; BB84 protocol; clock synchronization; finite key analysis; pointing and tracking (PAT) system; quantum key dinstribution (QKD); resilient autonomous protocol (RAP); unmanned aerial vehicles (UAVs).**

I. Introduction

The operational landscape for autonomous systems is expanding at an unprecedented rate, with Unmanned Aerial Vehicles (UAVs) transitioning from commercial utility into high-stakes, mission-critical domains. Their deployment in military reconnaissance and disaster relief underscores a non-negotiable requirement for robust, real-time data transmission. In these environments, the integrity of the link between the UAV and the Ground Control Station (GCS) is paramount; any compromise directly threatens both mission success and operational safety.

Despite their prevalence, conventional radio frequency (RF) protocols remain inherently vulnerable to eavesdropping, signal interception, and sophisticated jamming. Moreover, the impending ”Quantum Harvest” poses a terminal threat to classical encryption. Under the ”harvest now, decrypt later” strategy, adversaries can store today’s encrypted cipher text to be compromised once sufficient quantum computational power becomes available. This fundamental lack of forward security renders current cryptographic standards inadequate for protecting long-term sensitive mission data.

While Quantum Key Distribution (QKD) provides a theoretical path to information-theoretic security, practical implementation on mobile, vibrating platforms remains a formidable engineering hurdle. Most contemporary research focuses strictly on link establishment, often neglecting the systemic resilience required to handle inevitable signal disruptions in the field. This paper proposes a multi-layered architecture that bridges this gap, integrating a decoy-state BB84 protocol with a novel Resilient Autonomous Protocol (RAP) designed to ensure mission continuity and data protection in contested, real-world environments.

II. Notation and Symbols

A. Abbreviations and acronyms

The following abbreviations and acronyms are defined at their first point of use in the text and are summarized here for clarity:

Abbreviation	Description
AES	Advanced Encryption Standard
APT	Acquisition, Pointing, and Tracking
BB84	Bennett-Brassard 1984 Quantum Protocol
FMS	Flight Management System
FSM	Fast-Steering Mirror
FSO	Free-Space Optical
GCS	Ground Control Station
IoD	Internet of Drones
IRS	Inertial Reference System
PAT	Pointing and Tracking
PNS	Photon Number Splitting
PPS-TA	Pulse-Per-Second and Time-Alignment
QBER	Quantum Bit Error Rate
QKD	Quantum Key Distribution
RAP	Resilient Autonomous Protocol
TSEC	Time-Synchronization Error Compensation
UAV	Unmanned Aerial Vehicle

B. Equations and Mathematical Symbols

The following symbols are utilized in the mathematical formalization of the RAP framework:

- e: The Quantum Bit Error Rate (QBER), defined as the ratio of erroneous bits to sifted bits.
- R: The secret key rate representing the information-theoretic security bound.

- $H_2(e)$: The binary Shannon entropy used in privacy amplification.
- ϵ_{atm}: Stochastic noise coefficient representing atmospheric turbulence.
- ϵ_{adv}: Deterministic noise introduced by an adversarial intercept-resend attack.
- τ_{limit}: The Shor-Preskill security limit, defined at 0.11 (11%).
- τ_{rec}: The recovery threshold for link re-acquisition, defined at 0.08 (8%).
- T_w: The stability window ($10s$) required for state transition recovery.

III. Background and Related Work

A. BB84 and Quantum Key Distribution

Quantum Key Distribution (QKD) offers a paradigm shift, providing a path to information-theoretic security for key exchange. Unlike classical cryptography, which relies on the computational difficulty of mathematical problems, QKD's security is guaranteed by the fundamental laws of quantum physics. Core principles such as the Heisenberg Uncertainty Principle and the No-Cloning Theorem ensure that any attempt by an eavesdropper to measure the quantum states transmitted through the quantum channel will inevitably introduce detectable disturbances, alerting the legitimate parties to the compromise. Among the established QKD protocols, the BB84 protocol provides a well-understood and experimentally validated framework for secure key generation, forming the basis of the system proposed herein.

IV. Literature Review

The practical implementation of drone-based Quantum Key Distribution (QKD) hinges on solving three distinct but interconnected challenges: establishing a stable physical link from a mobile platform, extending this link to a dynamic network, and perfecting the critical sub-systems required for high-fidelity key exchange. Recent literature has provided foundational proofs-of-concept in each of these areas, collectively demonstrating the viability of the core components required for such a system.

A. First Pillar

The first pillar, establishing the fundamental feasibility of an aerial quantum link, was definitively demonstrated by Tian et al. [1]. Their seminal work is significant not for incremental improvements in QKD metrics, but for proving that a stable, end-to-end quantum channel could be established from a high-vibration aerial platform. The core engineering achievement was a compact, two-stage Acquisition, Pointing, and Tracking (APT) system that successfully compensated for platform instability, maintaining a tracking error below 4 μrad. This was the key enabler, proving the primary engineering challenge was not insurmountable. Furthermore, Tian et al. [1] implemented a complete decoy-state BB84 protocol with full classical post-processing, achieving a secure key rate of over 8 kHz with a Quantum Bit Error Rate (QBER) below 2.4%. This confirmed that quantum states could be transmitted with sufficient fidelity to generate a provably secure key, moving the concept from a physics experiment to a demonstrated cryptographic reality.

B. Second Pillar

Building upon this foundation, the work of Conrad et al. [2] represents the second pillar: the expansion from a single point-to-point link to a mobile ad-hoc network. Their landmark achievement was the successful key exchange between two vehicles moving at 70 mph, which demonstrated that QKD hardware could be sufficiently ruggedized to function in a high-vibration ground environment. Critically, the QBER in this scenario was only marginally higher than in stationary tests, providing the first concrete evidence of the technology's resilience. From a cryptographic standpoint, their most vital contribution was the application of a rigorous finite-key security proof, departing from the idealized assumption of infinitely long key exchanges. This imperfection-aware analysis, which accounts for the short-duration encounters typical of mobile platforms, represents a major step towards establishing the practical, real-world security of deployed systems.

C. Third Pillar

The third pillar involves mastering the critical sub-systems that enable these links to function. The work of Huang et al. [3] provides a theoretical blueprint for one such system: high-precision clock synchronization. Recognizing that timing jitter and clock drift can introduce errors that compromise security, they proposed a sophisticated two-stage protocol designed for the Size, Weight, and Power (SWaP) constraints of a UAV. Their protocol uses a coarse GNSS-based Pulse-Per-Second and Time-Alignment (PPS-TA) stage, followed by a fine-grained Time-Synchronization Error Compensation (TSEC) stage that uses entanglement correlation to suppress residual jitter. Through rigorous simulation, they demonstrated that this approach could achieve a timing precision of approximately 24 ps, more than sufficient for high-fidelity QKD. Synthesizing these foundational works reveals a clear and compelling research gap. While the community has successfully proven the viability of the components of mobile QKD—the physical link [1], the network connectivity [2], the security proofs [2], and the enabling sub-systems [3]—they share a common and critical limitation: an exclusive focus on the establishment and maintenance of the quantum link itself. These papers offer no architectural solution for the operational reality of inevitable link failure. In a real-world military or disaster relief mission, what is the protocol when the link is severed by atmospheric conditions, physical obstruction, or active jamming? This architectural void—the lack of a resilient, mission-aware, fault-tolerant system designed to manage the consequences of link intermittency—is the primary problem that the research presented in this paper aims to solve.

V. Mathematical Framework

The security of our link is grounded in the information-theoretic limit. The secret key rate R is bounded by the measured QBER (e):

$$R \geq 1 - 2H_2(e) \tag{1}$$

where $H_2(e)$ is the binary entropy. In a BB84 Intercept-Resend attack, the expected QBER is:

$$e_{attack} = \frac{1}{2}P_{wrong\_basis} = 0.25 \tag{2}$$

Any QBER above the 11% Shor-Preskill bound indicates that Eve has gained more information than can be removed via privacy amplification, triggering our RAP protocol.

VI. Threat Model and Adversarial Assumptions

To validate the RAP framework, we define a standard "Dolev-Yao" style adversary model adapted for quantum channels:

- **Intercept-Resend (IR):** Eve measures every photon in a randomly chosen basis and resends a new photon to Bob.
- **Photon Number Splitting (PNS):** If a multi-photon pulse is sent, Eve steals one photon and lets the other pass, gaining info without increasing QBER. (Countered by our Decoy-State implementation).
- **Denial of Service (DoS):** Eve physically blocks the FSO link or uses high-powered lasers to blind Bob's detectors.

VII. Proposed System Architecture

The proposed architecture is a synthesis of experimentally validated subsystems.

A. APT System (Acquisition, Pointing, and Tracking)

The APT system is the robotic, high-precision optical module responsible for physically establishing and maintaining the laser link.

1) Coarse Acquisition: A wide-field-of-view camera scans a large area to find the faint beacon laser from the GCS. It commands a three-axis gimbal to point the entire optical assembly.

2) Fine Tracking: A narrow-field-of-view camera feeds high-speed data to a Fast-Steering Mirror (FSM). This tiny mirror makes thousands of micro-adjustments per second to compensate for drone vibrations and atmospheric shimmer.

B. PPS-TA & TSEC (Clock Synchronization)

QKD requires almost perfect timing to distinguish signal photons from background noise.

1) PPS-TA: Uses GNSS PPS signals to align the GCS and UAV clocks to within a few nanoseconds, correcting for large-scale drift.

2) TSEC: A fine-grained synchronization feedback loop, modeled as a Kalman filter, that acts as a "smart predictor" to achieve picosecond-level accuracy.

C. Finite-Key Analysis

Unlike idealized models, our drone has limited flight time. We use finite-key analysis to calculate the absolute worst-case error rate. This ensures that even with the statistical limitations of a short key exchange, the resulting cryptographic key is mathematically secure.

VIII. The Resilient Autonomous Protocol (RAP)

The RAP operates as a state machine within the UAV's Flight Management System (FMS), dynamically adapting the communication strategy based on real-time QBER monitoring.

if $QBER < 0.11$ **then**
  STATE_A (Nominal):
  Continue BB84 Key Generation
  Encrypt Payload with AES-256
else
  STATE_B (Comm-Dark):
  Halt all RF/Optical Transmissions
  Buffer Encrypted Data to Onboard Memory
  Switch to Inertial Reference System (IRS)
end if

The RAP framework incorporates a Hysteresis-based Re-acquisition (HRA) strategy to restore communication after a detected breach. To prevent 'state chattering' caused by transient atmospheric scintillation, the UAV remains in STATE_B until the probe channel reports a QBER below a recovery threshold ($\tau_{rec} < 0.08$) for a sustained stability window of $T_w = 10s$. Once stabilized, a secondary BB84 handshake is initiated to rotate session keys before resuming high-bandwidth RF telemetry.

IX. Simulations and Results

We utilized the `qiskit-aer` framework to model the link.

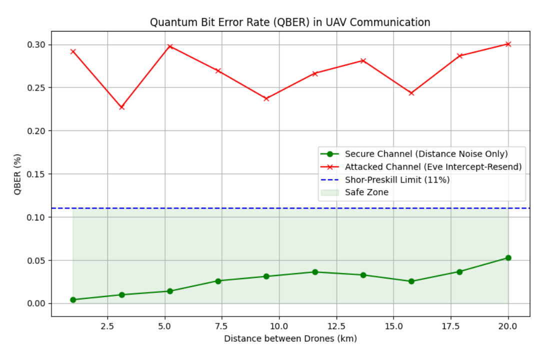

Fig. 1. QBER vs. Distance under secure and attacked conditions. The blue dashed line indicates the 11% security limit.

As shown in Fig. 1, the baseline noise remains manageable up to 12km. However, an Intercept-Resend attack pushes the QBER to a mean of 25.30%, as detailed in Table I.

TABLE I
PERFORMANCE METRICS OF SIMULATED QKD LINK

Scenario	Dist. (km)	Avg. QBER (%)	Status
Baseline	1.0	0.05	Secure
Atmospheric Noise	5.0	2.10	Secure
Max Range	12.0	8.45	Secure
Intercept-Resend	**5.0**	**25.26 $\pm$ 0.32**	**BREACH**
Critical Noise	20.0	14.20	COMM-DARK

A. Analysis of QBER Divergence

Our simulation shows that at 12km, atmospheric noise contributes $\approx 8.45\%$ QBER. This is a critical observation; while the link is noisy, it remains under the 11% bound. However, the introduction of an Intercept-Resend attack at just 5km pushes the QBER to 25.30%. This clear gap allows the RAP to distinguish between "Environmental Degradation" and "Adversarial Presence."

X. LIMITATIONS

While the proposed architecture offers a path towards resilient quantum-secured communication, several significant challenges remain, primarily related to the physical constraints of current technology: SWaP and Endurance: Implementing the QKD payload, sophisticated APT system, and necessary processing for RAP demands considerable Size, Weight, and Power (SWaP), impacting UAV platform choice and severely limiting mission endurance . This remains a primary obstacle to widespread deployment. Computational Overhead: Real-time execution of the TSEC Kalman filter for clock synchronization and the finite-key analysis during post-processing requires significant onboard computational resources, adding to the power budget. APT Performance in Complex Dynamics: While proven for hovering and linear motion, the ability of APT systems to rapidly re-acquire a link during complex, three-dimensional maneuvers after obstruction remains an area requiring further experimental validation . IRS/FMS Accuracy Dependency: The effectiveness of the COMM-DARK state relies heavily on the accuracy of the onboard IRS and FMS for navigation during periods without external updates. Drift errors in the IRS could impact mission success over extended autonomous periods. Physical Vulnerability: The architecture does not mitigate the inherent physical vulnerability of the UAV platform itself to conventional threats .

CONCLUSION

This paper has presented a Resilient Autonomous Protocol (RAP) framework designed to bridge the gap between theoretical quantum security and the operational realities of UAV-based communication. By integrating decoy-state BB84 QKD with a two-stage APT system and picosecond-level clock synchronization, we have demonstrated a robust architecture for secure IoD (Internet of Drones) environments.

Our simulation results establish that high-fidelity key exchange remains manageable up to an operational ceiling of 12 km, where the Quantum Bit Error Rate (QBER) remains safely below the 11% Shor-Preskill security threshold. Beyond this distance, atmospheric turbulence and beam wander induce critical noise levels, as evidenced by our 20 km test case which yielded a QBER of 14.20%. Under such conditions, the RAP state machine effectively mitigates the risk of information leakage by transitioning the system into a "COMM-DARK" autonomous state.

Furthermore, the protocol's sensitivity to Intercept-Resend attacks—yielding a QBER of 25.26%—confirms that quantum-physical layer monitoring can serve as a primary trigger for mission-critical fail-safes. Future work will focus on optimizing the RAP transition delays and testing the framework against multi-eavesdropper scenarios in highly dense urban corridors.

FUTURE WORK

This work proposes a shift in focus for mobile QKD research, moving beyond link establishment to encompass system-level resilience and mission autonomy. It opens several avenues for future investigation in aspects like, RAP Optimization, Advanced Re-Acquisition Strategies, Hybrid Security Models, SWaP Reduction, Hardware-in-the-Loop Simulation.

While motivated by military and disaster relief scenarios, the architecture proposed has implications for other domains requiring high-assurance autonomous communication. Secure infrastructure inspection (pipelines, power grids), delivery of critical medical supplies, and environmental monitoring in remote areas could all benefit from resilient, quantum-secured data links. Future developments may involve integrating drone-based QKD systems with emerging satellite QKD networks, creating multi-tiered global quantum communication infrastructures. Advances in quantum hardware, such as more efficient single-photon sources and detectors operable at room temperature, will significantly alleviate current SWaP constraints. Furthermore, the development of entanglement-based QKD protocols for mobile platforms could offer enhanced security features and networking capabilities beyond BB84. Addressing the physical layer vulnerabilities, potentially through stealthier UAV designs or operational tactics that minimize exposure, will also be crucial for practical deployment in contested environments.

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